Experiment 2: Exploring Packet Sniffer and Packet Analyzer.

Question 1.

what are the different protocols you observe at the following layers of the protocols stack?

A. Application layer
B. Transport layer
C. Network Layer

→ Protocols observed at the application layer are :-

(i) HTTP (Hypertext transfer protocol)
(ii) DNS (Domain Name System)
(iii) SSDP (Simple Service Discovery protocol)

→ At Transport layer are:-

(i) TCP (Transmission Control protocol)
(ii) UDP (User Datagram protocol)
(iii) TLS (Transport layer Security)

→ At Network layer protocols are:-

(i) IPv4 (Internet protocol version 4)
(ii) IPv6 (Internet protocol version 6)
(iii) ARP (Address Resolution Protocol)

Question 2.

what is the total amount of data being received for the following two cases?

A. when you access http://neverssl.com/
B. when you access https://www.cornell.edu

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	14071	100.0	14155544	718 k	0	0	0	14071
Ethernet	100.0	14071	1.4	196994	10 k	0	0	0	14071
Internet Protocol Version 4	99.9	14053	2.0	281060	14 k	0	0	0	14053
User Datagram Protocol	1.0	135	0.0	1080	54	0	0	0	135
Simple Service Discovery Protocol	0.0	4	0.0	680	34	4	680	34	4
QUIC IETF	0.5	65	0.2	34499	1751	65	33067	1679	70
Multicast Domain Name System	0.0	6	0.0	776	39	6	776	39	6
Domain Name System	0.4	60	0.0	4953	251	60	4953	251	60
Transmission Control Protocol	98.9	13918	2.0	288940	14 k	13143	273440	13 k	13918
Transport Layer Security	1.7	235	1.7	236615	12 k	235	214784	10 k	242
Hypertext Transfer Protocol	3.7	526	92.3	13064215	663 k	503	520834	26 k	526
Portable Network Graphics	0.0	1	0.0	124	6	1	124	6	1
Media Type	0.1	21	88.5	12532075	636 k	21	12532075	636 k	21
Line-based text data	0.0	1	0.0	300	15	1	300	15	1
Data	0.1	14	0.0	14	0	14	14	0	14
Address Resolution Protocol	0.1	18	0.0	504	25	18	504	25	18

→ While accessing 1st, I received data of 848bytes. (4 packets)
And while accessing 2nd, I received data of 14 MB. (10,890 packets).

Question 3.

A. How many HTTP GET requests do you observe? List down the GET requests.

→ 3 HTTP GET requests.

1. /online/
2. /online
3. /favicon.ico

B. What version of HTTP the server is running?

→ HTTP/1.1.

C.(i) The total number of TCP segments being received:

→ 7.

(ii) The total amount of data being received in corresponding HTTP response message.

→ Seq = 12643, Ack = 4456, Win = 61440

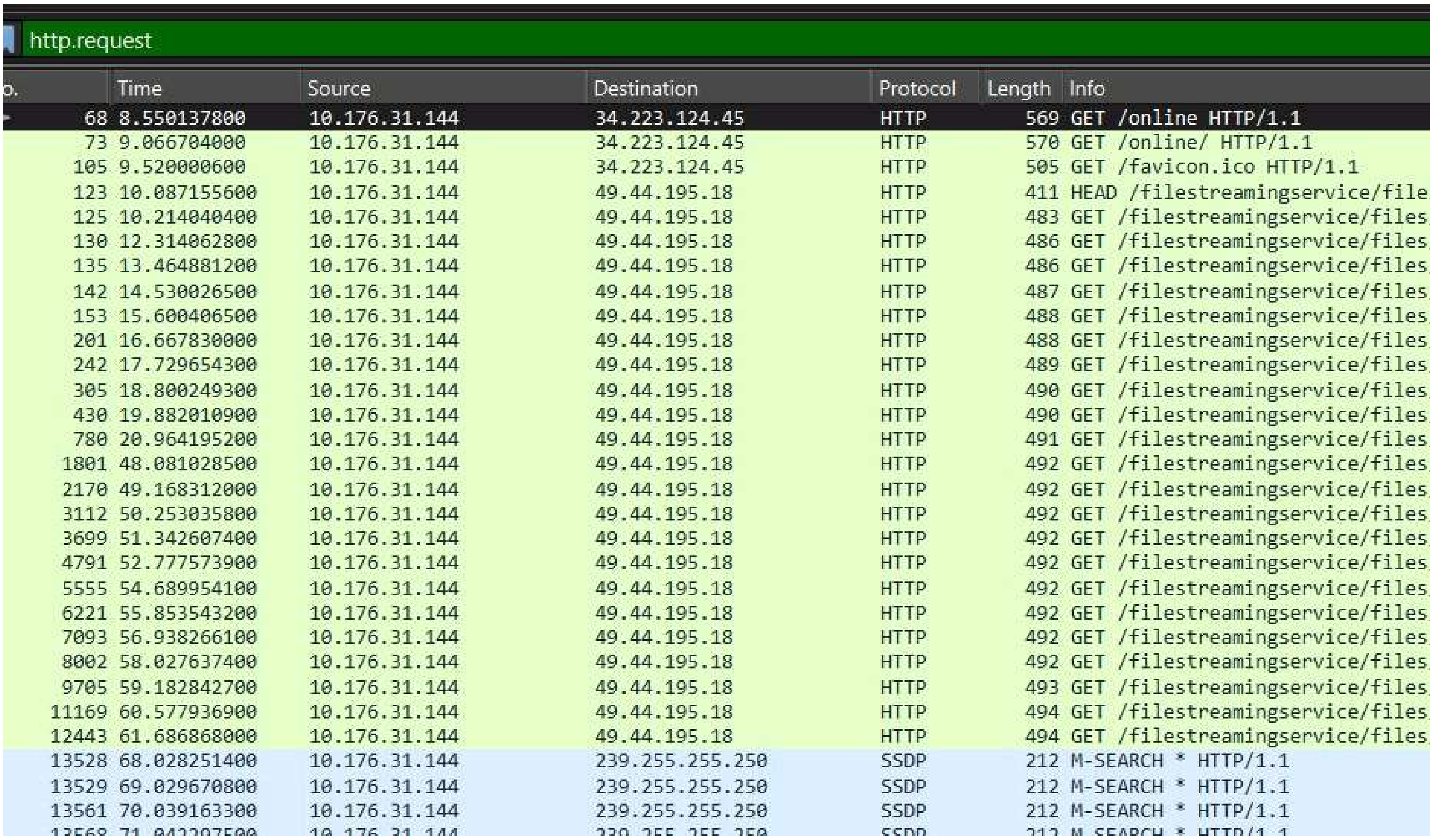
D.

(i) SYN → The process two computers used to establish reliable connection before transferring.

(ii) SYN-ACK → Establish a reliable error-free connection between client & server.

(iii) ACK → fundamental Networking process.

Address A	Address B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
8e:bc:bf:e8:e2:28	ff:ff:ff:ff:ff:ff	2	84 bytes	1	2	84 bytes	0	0 bytes	49.774306800	60.047815	11 bits/s	0 bits/s
94:bb:43:82:86:f6	01:00:5e:00:00:fb	6	1 kB	2	6	1 kB	0	0 bytes	64.137864300	0.775612	10 kbps	0 bits/s
94:bb:43:82:86:f6	01:00:5e:7f:ff:fa	4	848 bytes	3	4	848 bytes	0	0 bytes	68.028251400	3.014046	2250 bits/s	0 bits/s
94:bb:43:82:86:f6	7a:15:34:eb:fb:cf	14,059	14 MB	0	3,169	316 kB	10,890	14 MB	0.000000000	157.544578	16 kbps	702 kbps



http.request

o.	Time	Source	Destination	Protocol	Length	Info
68	8.550137800	10.176.31.144	34.223.124.45	HTTP	569	GET /online HTTP/1.1
73	9.066704000	10.176.31.144	34.223.124.45	HTTP	570	GET /online/ HTTP/1.1
105	9.520000600	10.176.31.144	34.223.124.45	HTTP	505	GET /favicon.ico HTTP/1.1
123	10.087155600	10.176.31.144	49.44.195.18	HTTP	411	HEAD /filestreamingservice/file
125	10.214040400	10.176.31.144	49.44.195.18	HTTP	483	GET /filestreamingservice/files
130	12.314062800	10.176.31.144	49.44.195.18	HTTP	486	GET /filestreamingservice/files
135	13.464881200	10.176.31.144	49.44.195.18	HTTP	486	GET /filestreamingservice/files
142	14.530026500	10.176.31.144	49.44.195.18	HTTP	487	GET /filestreamingservice/files
153	15.600406500	10.176.31.144	49.44.195.18	HTTP	488	GET /filestreamingservice/files
201	16.667830000	10.176.31.144	49.44.195.18	HTTP	488	GET /filestreamingservice/files
242	17.729654300	10.176.31.144	49.44.195.18	HTTP	489	GET /filestreamingservice/files
305	18.800249300	10.176.31.144	49.44.195.18	HTTP	490	GET /filestreamingservice/files
430	19.882010900	10.176.31.144	49.44.195.18	HTTP	490	GET /filestreamingservice/files
780	20.964195200	10.176.31.144	49.44.195.18	HTTP	491	GET /filestreamingservice/files
1801	48.081028500	10.176.31.144	49.44.195.18	HTTP	492	GET /filestreamingservice/files
2170	49.168312000	10.176.31.144	49.44.195.18	HTTP	492	GET /filestreamingservice/files
3112	50.253035800	10.176.31.144	49.44.195.18	HTTP	492	GET /filestreamingservice/files
3699	51.342607400	10.176.31.144	49.44.195.18	HTTP	492	GET /filestreamingservice/files
4791	52.777573900	10.176.31.144	49.44.195.18	HTTP	492	GET /filestreamingservice/files
5555	54.689954100	10.176.31.144	49.44.195.18	HTTP	492	GET /filestreamingservice/files
6221	55.853543200	10.176.31.144	49.44.195.18	HTTP	492	GET /filestreamingservice/files
7093	56.938266100	10.176.31.144	49.44.195.18	HTTP	492	GET /filestreamingservice/files
8002	58.027637400	10.176.31.144	49.44.195.18	HTTP	492	GET /filestreamingservice/files
9705	59.182842700	10.176.31.144	49.44.195.18	HTTP	493	GET /filestreamingservice/files
11169	60.577936900	10.176.31.144	49.44.195.18	HTTP	494	GET /filestreamingservice/files
12443	61.686868000	10.176.31.144	49.44.195.18	HTTP	494	GET /filestreamingservice/files
13528	68.028251400	10.176.31.144	239.255.255.250	SSDP	212	M-SEARCH * HTTP/1.1
13529	69.029670800	10.176.31.144	239.255.255.250	SSDP	212	M-SEARCH * HTTP/1.1
13561	70.039163300	10.176.31.144	239.255.255.250	SSDP	212	M-SEARCH * HTTP/1.1